

Quantum theory of radiation by nonstationary systems with application to high-order harmonic generation (Phys. Rev. A 101, 013410 (2020))

Statement of the main problem

- We are now interested in solving the following coupled ode system

$$\begin{cases} i \frac{\partial}{\partial t} \psi(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \psi(x, t), \\ i \frac{\partial}{\partial t} \phi_\omega(x, t) &= \left(\hat{H}_a + \hat{V}_{\text{ext}}(t) \right) \phi_\omega(x, t) + e^{i\omega t} \hat{p} \psi(x, t), \end{cases} \quad (1)$$

- So we want our drivers to solve the following general case as efficiently as possible

$$i \partial_t \phi_\omega(x, t) = H(t) \phi_\omega(x, t) + S(t), \quad H(t) = H_0 + F(t)x, \quad (2)$$

- Considering, the solution to the homogeneous problem, without the source

$$i \partial_t U(t, t_0) = (H_0 + F(t)x) U(t, t_0), \quad U(t, t_0) = \mathbb{1}, \quad (3)$$

we can then, from Duhamel principle write

$$\partial_t \phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t) - i \int_{t_0}^{t_1} U(t - \tau) S(\tau) d\tau. \quad (4)$$

The above is the exact formulation of our problem.

Geometric operators

Norm-preserving propagation schemes

$$\partial_t \phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t) - i \int_{t_0}^{t_1} U(t - \tau) S(\tau) d\tau. \quad (4)$$

- Our test system will be a forced QHO, with source

$$\begin{cases} i\partial_t \psi(x, t) &= \left(-(1/2)\partial_x^2 + (1/2)x^2 + F(t)x \right) \psi(x, t), \\ i\partial_t \phi_\omega(x, t) &= \left(-(1/2)\partial_x^2 + (1/2)x^2 + F(t)x \right) \phi_\omega(x, t) - ie^{i\omega t} \partial_x \psi(x, t), \end{cases} \quad (\hbar = \omega = m = 1) \quad (5)$$

- The evaluation of both the terms shall be done with matching order methods
- We will try the following geometric (norm-preserving) propagators

- Midpoint Duhamel (order 2)
- Yoshida scheme (order 4)
- Runge-Kutta 4 (EFRK4)?
- etc...

Geometric operators

Convergence criteria

$$\partial_t y = Ay + S(x, t). \quad (6)$$

- Assessment of these schemes should be done based on two different tests with the number of steps N , and the time step h are such as

$$N_\ell = N_0 2^\ell, \quad h_\ell = \frac{T}{N_0 2^\ell}. \quad (7)$$

- Absolute convergence

$$p_\ell^{\text{abs}} = \frac{\log(E_{\ell-1}^{\text{abs}}/E_\ell^{\text{abs}})}{\log(2)}, \quad \text{with} \quad E_\ell^{\text{abs}} = |y_\ell(T) - y_{\text{exact}}(T)|. \quad (8)$$

- Self-consistency

$$p_\ell^{\text{self}} = \frac{\log(D_{\ell-1}/D_\ell)}{\log(2)} \quad \text{with} \quad D_\ell = |y_\ell(T) - y_{\ell-1}(T)|, \quad D_\ell \sim Ch_\ell^p. \quad (9)$$

- However if we have to fix the test case to the inhomogeneous Schrödinger equation, then we can realistically only carry the second one here

Geometric operators

Midpoint Duhamel - 2nd order approximation

$$\partial_t \phi_\omega(x, t) = U(t, t_0) \phi_\omega(x, t) - i \int_{t_0}^{t_1} U(t - \tau) S(\tau) d\tau. \quad (4)$$

- We can show that the above can be expressed as

$$\phi_{n+1}^\omega = U(\Delta t) \phi_n^\omega - i \Delta t U(\Delta t/2) S_{n+1/2}. \quad (10)$$

where the source is evaluated at $t + \Delta t/2$, i.e

$$S_{n+1/2} = e^{i\omega(t+\Delta t/2)} \partial_x \psi_{n+1/2}, \quad \text{with} \quad \psi_{n+1/2} = U(\Delta t/2) \psi_n. \quad (11)$$

This is the midpoint Duhamel formula, of order 2.

- The driver shall be regarded as decent if the expected order is met

Geometric operators

Midpoint Duhamel - Results

$$\phi_{n+1}^\omega = U(\Delta t)\phi_n^\omega - i\Delta t U(\Delta t/2) S_{n+1/2}. \quad (10)$$

$$S_{n+1/2} = e^{i\omega(t+\Delta t/2)} \partial_x \psi_{n+1/2}, \quad \text{with} \quad \psi_{n+1/2} = U(\Delta t/2)\psi_n. \quad (11)$$

t	$\Re \psi(1)$	$\Im \psi(1)$	$\ \psi\ $	$\Re \phi(1)$	$\Im \phi(1)$	$\ \phi\ $
-1.0	1.0000	-0.0000	1.0000	0.0574	-0.0000	0.6248
-0.9	0.9979	-0.0482	1.0000	0.0640	-0.0018	0.6432
-0.5	0.9377	-0.2263	1.0000	0.1193	-0.0066	0.7814
-0.1	0.7927	-0.3337	1.0000	0.1885	-0.0195	0.8795
0.0	0.7484	-0.3437	1.0000	0.2036	-0.0192	0.8823
0.5	0.5428	-0.2995	1.0000	0.2925	0.0198	0.8257
0.9	0.4566	-0.2195	1.0000	0.3965	0.0396	0.8201
1.0	0.4474	-0.2054	1.0000	0.4222	0.0347	0.8347

Refinement level k	Time step Δt	$ \psi^{(k)} - \psi^{(k+1)} $	$ \phi^{(k)} - \phi^{(k+1)} $	Observed order
1	.100	3.6596×10^{-6}	2.7621×10^{-5}	—
2	.050	9.1488×10^{-7}	4.9953×10^{-6}	—
3	.025	2.2872×10^{-7}	1.2522×10^{-6}	$p_\psi = 2.0000$ $p_\phi = 1.9961$

- Results of the time propagation for the coupled forced QHO with and without source (Eq.5)

Ne = 2
Np = 6
-xmin = xrange/2 = 7
-t0 = duration/2 = 1

Initial conditions
 $\psi_0(x) = (\pi)^{-1/4} e^{-(x-x_c)^2/2}$
 $\phi_0^\omega(x) = \partial_x \psi_0(x)$
 $x_c = 0$

$$p_k^{\text{self}} = (\text{err}_{k-1}/\text{err}_k) / \log(2)$$